Acoustic and EMA Data Fusion at the phonetic level for Dysarthric Speech Recognition: A Study Using the TORGO Database

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Abstract

Instead of using standard speech samples for dysarthric speech recognition, we have a hypothesis that by breaking down the speech samples into phones, the dysarthric phone will be much more similar to a typical speaker. Based on this hypothesis, we have two research questions: RQ1) How does phone-based acoustic data fare in comparison to word-level acoustic data? RQ2) Does phone-based electromagnetic articulograph data generalize well when compared with phone-based acoustic data? We used TORGO data for our research and found that the phone-based data does generalize well when compared with the entire speech sample for dysarthric speakers. Moreover, the EMA data were much more similar to controlled and other levels of dysarthric speakers in capturing the phone [f].

Index Terms: dysarthric speech recognition, phonetics.

1. Introduction

The current ASR models, such as Wave2Vec 2.0 from Meta [1] and Whisper from OpenAI [2], are proficient in recognizing typical speech in multiple languages. However, their performance on impaired speech remains consistent. Various efforts have been made to enhance dysarthric speech recognition, including augmenting data from typical speech to dysarthric speech [3] for building personalized ASR models [4], adapting acoustic [5] and lexical models to improve dysarthric speech [6], incorporating articulograph data to enhance dysarthric speech recognition [7], utilizing multiple databases to improve dysarthric speech recognition [8], and employing large deep learning models [9][10] to enhance ASR systems.

In this work, we explore the use of phone-based acoustic and articulograph data instead of using the entire speech sample for training. We utilize the well-known TORGO database [11] for our research. To assess the effectiveness of phone-based data, we employ classic Dynamic Time Warping [12] to compare dysarthric speakers against both other dysarthric and controlled speakers. We conduct two experiments: in the first experiment, the sliced phone sample is compared with the entire speech sample to demonstrate the effectiveness of phone-based sliced data. In the second experiment, sliced phone electromagnetic articulography (EMA) data is compared against sliced phone acoustic data. The results show that phone-based data is much more similar to other speakers than using the entire word sample. Additionally, when using sliced EMA data to compare against other speakers, certain phones like the fricative [f] are much more similar to other speakers than the phone-level acoustic data.

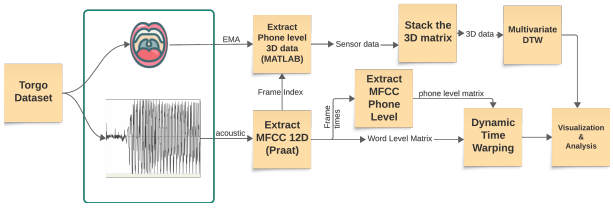


Figure 1: *Domain Driven Diagram*

2. Experiments

Approach: The phones form the individual components of speech. The problem with dysarthric speech recognition is due to the inter and intra variation in dysarthric speakers. The inter variation is due to the differences from typical speech in terms of vowel duration [11] and lot of pauses compared to typical speakers [13]. The intra variation arises due to the severity levels (mild, moderate, and severe) in dysarthria. By breaking down the speech into individual phone units, the dysarthric phone can be much more similar to typical speakers' phones and also to other dysarthric-level speakers. In order to test the above hypothesis, we are carrying out two experiments. **Task:** The first experiment consists of comparing phone-level acoustic data of dysarthric speakers against controlled and other dysarthric speakers using dynamic time warping, keeping word-level acoustic data as the baseline. In experiment two, phone-level Electromagnetic Articulography (EMA) data of dysarthric speakers are compared against controlled and dysarthric speakers to check for alignment cost.

2.1. Phone Level Extraction, Parsing and Storing

The sound files and the EMA files for all the speakers are processed in four stages. The first stage involves the extraction of Mel Frequency Cepstral Coefficient (MFCC) matrices for all the speakers' .wav files. Simultaneously, the start and end values of phone boundaries are also stored in the .txt file for every speaker's prompt during the first stage. The second stage involves parsing the .txt file to extract the start and end time phone boundaries for every prompt and then slicing the matrix related to the prompt accordingly. Now, the third stage is to extract the phone-level matrix for the EMA data by applying the start and end time values from the parsed .txt files using MATLAB. The final stage is to create a JSON data structure where all the prompts, along with the matrix and EMA sliced path, are stored for easy reference. Up to this stage, the process is the same for RQ1 and RQ2, as shown in Figure 1.

2.2. Comparison of Acoustic Data: Word Level vs. Phone Level

In RQ1, the JSON data structure is parsed to obtain both the word-level matrix and the phone-level matrix for the same prompt across all speakers. For each dysarthric speaker $\mathcal{M}d$, word-level and phone-level acoustic data are compared against $\mathcal{M}c$ and the remaining $\mathcal{M}d$ speaker using the Dynamic Time Warping algorithm (DTW) [12]. The word-level acoustic data is used as the baseline here.

2.3. Comparison in Phone-Level: EMA vs. Acoustic

Similarly, in RQ2, the JSON data structure is parsed again to obtain phone-level EMA data and acoustic data for all the speakers. In this stage, the phone-level 3D EMA data for the sensors Tongue back, Tongue middle, Tongue tip, Upper lip, Lower lip, Left lip, and Right lip are stacked horizontally, leaving out the remaining sensors as they are solely for head correction. Then, for each dysarthric speaker $\mathcal{M}d$, phone-level acoustic and EMA data are compared against $\mathcal{M}c$ and the remaining $\mathcal{M}d$ speakers using multivariate dynamic time warping (MDTW) [14]. The phone-level acoustic data is used as a baseline here.

3. Dataset

The TORGO database [11] consist of aligned acoustic and measured 3D articulatory speech from both typical (controlled) and dysarthric English speakers. The dysarthric speakers either have cerebral palsy (CP) or amyotrophic lateral sclerosis (ALS) which are the two most prevalent cause of disability [15]. The female speakers $\mathcal{F}s$ were excluded from the dataset due to the absence of articulatory and phonetic transcription files for a substantial number of controlled $\mathcal{F}c$ and dysarthric $\mathcal{F}d$ female speakers. Additionally, the male controlled $\mathcal{M}c$ and dysarthric $\mathcal{M}_d$ data were further refined by removing empty and unclear prompts. The total utterances in the refined TORGO data was 5801, Table 1 describes the distribution of prompts for each male speakers including their dysarthric level. The dysarthric level for $\mathcal{M}02$, $\mathcal{M}03$, $\mathcal{M}04$ and $\mathcal{M}05$ was labeled based on the SLP label provided in the [16] for TORGO dataset. The $\mathcal{M}01$ dysarthric level was chosen based on the experiments performed in [17].

Table 1: *Cleaned TORGO data*

Speakers	Dysarthric level	Utterances
$\mathcal{MC}01$	typical	1094
$\mathcal{MC}02$	typical	690
$\mathcal{MC}03$	typical	886
$\mathcal{MC}04$	typical	1011
$\mathcal{M}01$	severe	382
$\mathcal{M}02$	moderate	400
$\mathcal{M}03$	typical	413
$\mathcal{M}04$	severe	414
$\mathcal{M}05$	severe	512

3.1. Data Exploration

The total number of unique phones available in the time-aligned phonetic transcription for $\mathcal{M}_c$ and $\mathcal{M}_d$ is 84. Out of the 84 phones based on CMU pronouncing dictionary[1] phoneme set we divided the 84 phones into 24 consonants and 15 vowels, the other 45 phones consist of misspelled phones such as ([se], [j], [kc] , etc.) and 8 silence phones such as ([bcl], [kcl], [sil], etc.). The 3D Electromagnetic articulography data for each speakers prompt has the following 3D matrix shape (NumPoints, 7, 12), the third dimension (12) channels point to sensors used for measuring the inside and outside movement of the vocal tract. The channels in these data refer to the following positions in the vocal tract:

1. Tongue back (TB)
2. Tongue middle (TM)
3. Tongue tip (TT)
4. Forehead
5. Bridge of the nose (BN)
6. Upper lip (UL)
7. Lower lip (LL)
8. Lower incisor (LI)
9. Left lip
10. Right lip
11. Left ear
12. Right ear

The second dimension (7) refers to the following columns x, y, z, phi, theta, rms, extra, where x,y and z are the co-ordinates, phi and theta are angles to the spherical coordinates, rms is the root mean squared error recorded in AG500 instrument and extra are always zero in this case. The first dimension Numpoints can take any number of values depending upon the length of the time. Each row in (Numpoints, 7) will have $\frac{1}{200}$th second of the sound file for all the channels.

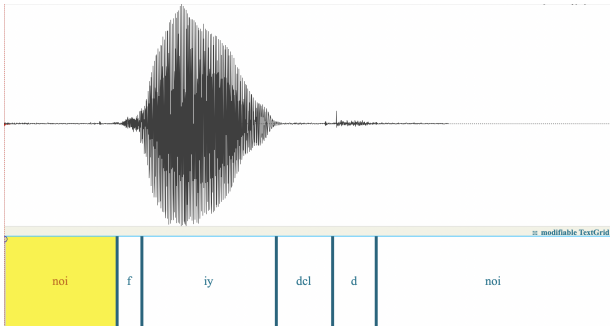
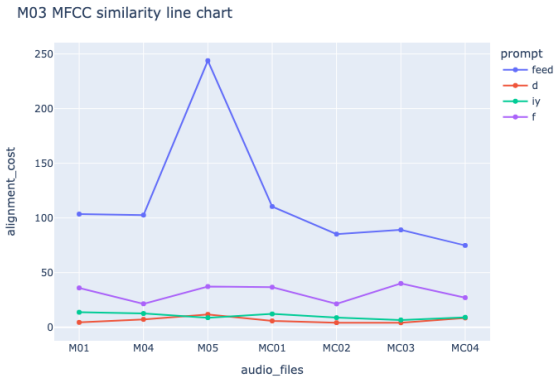


Figure 2: *Visualization of segmentation and annotation in Praat tool*

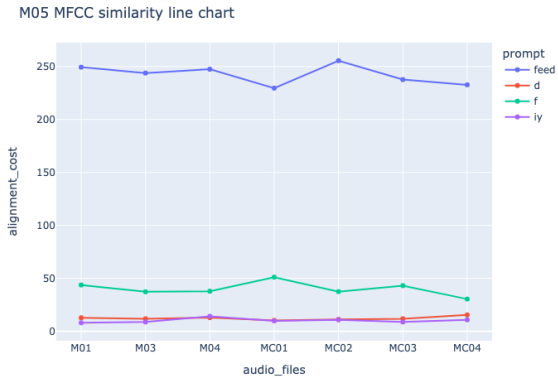
3.2. Segmentation & Annotation

The TORGO dataset already has orthographic and phonetic transcription (ARPAbet) for every speakers prompt. But not all the segmentation for the phonetic transcriptions are accurate. For few of the transcriptions the phone boundary is drawn completely wrong matching with a different sound. In the boundary mismatch cases the segmentation was done manually again using a software tool called Praat [18] to align the boundaries with their respective phone sounds. An illustration depicting the segmentation and annotation performed using the Praat tool is presented in Figure 2

[1]CMU Pronouncing dictionary, `http://www.speech.cs.cmu.edu/cgi-bin/cmudict`

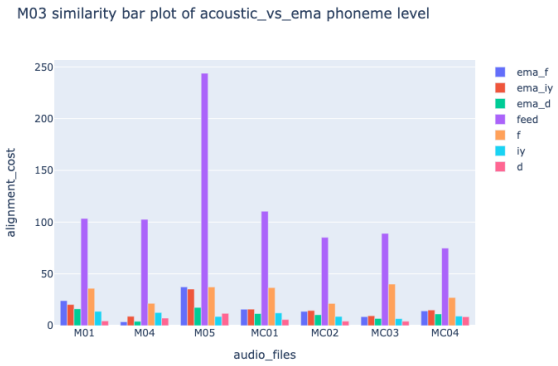


(a) *M03 Acoustic comparison of word level vs phone level*

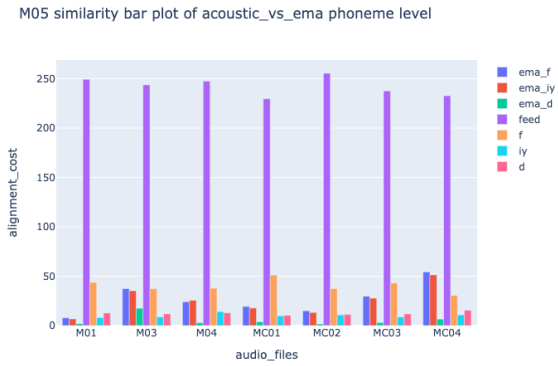


(b) *M05 Acoustic comparison of word level vs phone level*

Figure 3: *Acoustic comparison of word level vs phone level for M03 and M05*



(a) *M03 Acoustic comparison of word level vs phone level*



(b) *M05 Acoustic comparison of word level vs phone level*

Figure 4: *Acoustic vs. EMA phone level comparison for M03 and M05*

4. Results

Evaluation Metrics: The alignment cost is the chosen evaluation metric for both RQ1 and RQ2. The similarity between two speakers is lower if the alignment cost is higher, and vice versa.

4.1. RQ1 Word-level vs. Phone level

In Figure 3, we are comparing the dysarthric speakers $\mathcal{M}03$ in Figure 3a and $\mathcal{M}05$ in Figure 3b against all the other speakers. We analyze whether the sliced acoustic data like [f], [iy], [d] generalize well against other speakers when compared to the phoneme-based [fiyd] acoustic data. This comparison is made against $\mathcal{M}c$ speakers, including other $\mathcal{M}d$ speakers.

4.2. RQ2 Acoustic vs EMA in Phone level

Similarly, in Figures 4a and 4b, we are comparing EMA phone-level data against acoustic phone-level data. The sliced phones [f], [iy], [d] are used to calculate the similarity for $\mathcal{M}03$ and $\mathcal{M}05$ against $\mathcal{M}c$ speakers, including other $\mathcal{M}d$ speakers. The phoneme [fiyd] is used as a baseline here to demonstrate the differences in alignment cost.

5. Discussion

Acoustic dysarthric phone is similar to typical speakers phone. As shown in Figure 3, the alignment cost for comparing dysarthric speakers at the word level is remarkably high, indicating dissimilarity. However, when the same word is broken down into the phone level, the alignment costs are much lower, closer to the x-axis, showing high similarity. Astonishingly, for the severely affected speaker $\mathcal{M}05$, as shown in Figure 3b, the phone-level similarity was much higher, indicating that dysarthric MFCC phone-level data can be used for training due to its high similarity with typical and other dysarthric speakers.

Articulograph Dysarthric phone is even more similar to typical speakers phone. For the fricative phone [f], the phone-level EMA data was almost similar when compared to other speakers in Figure 4, indicating the usefulness of the articulograph data. However, in other cases like [iy] and [d], the acoustic phone-level similarity was much closer to other speakers. This opens up the research field to investigate what phones EMA can capture better than acoustic phones.

Limitations and Future Work: The data used for the analysis was only one prompt ”feed”. This is not enough to conclude that phone-level data are much similar to typical phones.

A large number of prompts need to be taken for each place of articulation to generalize the data, and a statistical analysis has to be conducted to understand the similarity of dysarthric phones to typical phones. Based on the research we can do fusion on EMA phone level data and acoustic phone level data to generalize the dysarthric speech well with other speakers.

6. Conclusion

In this work, we compared the dysarthric speakers' phone-level acoustic and articulograph data against other speakers using dynamic time warping. The dysarthric fricative phones like [f] are highly similar to other speakers when using articulograph data, but in other cases, such as [iy] and [d], the acoustic phone-level data are highly similar to other speakers.

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